

1.3 Computer networks, connections and protocols

Where this unit sits, what it covers, and how far

Where this unit sits

Qualification	GCSE Computer Science (9-1), specification code J277, from 2020
Component	Component 01: Computer systems — 80 marks, 1 hour 30 minutes
What OCR says the component covers	the central processing unit (CPU), computer memory and storage, data representation, wired and wireless networks, network topologies, system security and system software
This unit	topic 1.3 Computer networks, connections and protocols
Checked	3 October 2026, at https://www.ocr.org.uk/qualifications/gcse/computer-science-j277-from-2020/specification-at-a-glance/

How far this has been checked. The topic numbers and titles below are the ones this course uses on its worksheets and slides. They were not re-checked against the published specification document when this map was built: the document could not be retrieved from this build environment. Check them against <https://www.ocr.org.uk/Images/558027-specification-gcse-computer-science-j277.pdf> before relying on this map for a scheme of work, and treat any finer subdivision as absent rather than assumed - no sub-point numbers are claimed anywhere in these documents, because none were verified.

Before it: 1.1 Systems architecture, 1.2 Memory and storage

After it: 1.4 Network security, 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments, PY Python programming

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Types of network: zoom out nw-101	stretch work, no new content	27	35
2	Network control room nw-102	stretch work, no new content	14	23
3	Network showdown nw-103	What are the differences between peer-to-peer and client-server networks?	13	26
4	Mission: wire up the school nw-104	stretch work, no new content	13	16
5	Inside the internet nw-105	How does the internet work?	13	27
6	Star and mesh networks nw-106	stretch work, no new content	22	32

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
7	Revision HQ nw-107	stretch work, no new content	19	23
8	Assessment 1 nw-108-test	assesses the whole unit	0	0
9	Connection café nw-109	Which is better: a wired or wireless network?	13	28
10	Cipher room nw-110	What is the purpose of encryption?	13	24
11	Address sorting office nw-111	What are the differences between the three types of network address?	13	21
12	Protocol city nw-112	What are standards and protocols?	14	28
13	Parcel depot nw-113	What are the benefits of layering protocols?	12	25
14	Assessment 2 reflection nw-114-test	assesses the whole unit	0	0
14	Revision HQ 2 nw-114	stretch work, no new content	20	30

Prior knowledge, vocabulary and skills

Assumed before the unit: the outcomes of 1.1, 1.2

Vocabulary introduced: Client-server, Peer-to-peer, DNS (Domain Name Service), Hosting, The cloud, Ethernet, Wi-Fi, Bluetooth, Encryption, Public key, Private key, MAC address, IP address, Standard, Protocol, Layer, IPv4, IPv6, MAC, Protocols, Layers

What a pupil does, not only reads: match, mcq, multi, order, sort

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage.json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know what is meant by a standalone computer nw-101-o1	nw-l01	yes	yes	yes	Full
know the different types of network: LAN and WAN nw-101-o2	nw-l01	yes	yes	yes	Full
understand the advantages of networking nw-101-o3	nw-l01	yes	yes	yes	Full
understand the implications (disadvantages) of networking nw-101-o4	nw-l01	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand what factors can affect the performance of a network nw-102-o1	nw-102	yes	yes	yes	Full
be able to explain how each factor affects network speed nw-102-o2	nw-102	yes	yes	yes	Full
know what a client-server model is nw-103-o1	nw-103	yes	yes	yes	Full
know what a peer-to-peer model is nw-103-o2	nw-103	yes	yes	yes	Full
understand the different roles computers have in each model nw-103-o3	nw-103	yes	yes	yes	Full
know the hardware needed to connect a LAN nw-104-o1	nw-104	yes	yes	yes	Full
understand the purpose of each piece of hardware and the jobs it does nw-104-o2	nw-104	—	—	yes	Partial
understand what the internet actually is nw-105-o1	nw-105	—	—	yes	Partial
understand the terms Domain Name Service and domain name server nw-105-o2	nw-105	yes	yes	yes	Full
understand what is meant by hosting and the cloud nw-105-o3	nw-105	yes	yes	yes	Full
understand what is meant by the terms web server and client nw-105-o4	nw-105	yes	yes	yes	Full
know what a star network is nw-106-o1	nw-106	yes	yes	yes	Full
know what a mesh network is nw-106-o2	nw-106	yes	yes	yes	Full
understand that the internet is an example of a partial mesh network nw-106-o3	nw-106	yes	yes	yes	Full
know the advantages and disadvantages of star and mesh networks nw-106-o4	nw-106	yes	yes	yes	Full
know which lessons I am secure on and which I need to revise nw-107-o1	nw-107	yes	yes	yes	Full
have completed any outstanding work from lessons 1 to 6 nw-107-o2	nw-107	yes	yes	yes	Full
be ready for my mid-topic test next lesson nw-107-o3	nw-107	yes	yes	—	Partial
understand that Ethernet is a wired method of connection nw-109-o1	nw-109	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
understand that Wi-Fi and Bluetooth are wireless methods of connection nw-l09-o2	nw-l09	yes	yes	yes	Full
understand the benefits and drawbacks of wired versus wireless connections nw-l09-o3	nw-l09	yes	yes	yes	Full
be able to recommend a connection type for a specified scenario nw-l09-o4	nw-l09	yes	yes	yes	Full
know the basics of how cryptography can work with a simple key nw-l10-o1	nw-l10	yes	yes	—	Partial
know how wireless devices authenticate each other before communicating data nw-l10-o2	nw-l10	yes	yes	—	Partial
understand the difference between private and public keys nw-l10-o3	nw-l10	yes	yes	yes	Full
understand why private keys are never shared nw-l10-o4	nw-l10	yes	yes	yes	Full
understand the uses of MAC and IP addressing nw-l11-o1	nw-l11	yes	yes	yes	Full
understand the difference between IPv4 and IPv6 nw-l11-o2	nw-l11	yes	yes	yes	Full
understand the need for IPv6 nw-l11-o3	nw-l11	yes	yes	yes	Full
understand the need for standards in computing nw-l12-o1	nw-l12	yes	yes	yes	Full
understand the seven common protocols and what they are used for nw-l12-o2	nw-l12	yes	yes	yes	Full
know why protocols are layered nw-l13-o1	nw-l13	yes	yes	yes	Full
understand the advantages of layering nw-l13-o2	nw-l13	yes	yes	yes	Full
know which lessons I am secure on and which to revise nw-l14-o1	nw-l14	yes	yes	—	Partial
be ready for the end-of-unit assessment nw-l14-o2	nw-l14	—	—	—	Not covered

32 full · 6 partial · 1 not covered, of 39 outcomes.

Gaps, assumptions and overlap

- **nw-l04-o2** understand the purpose of each piece of hardware and the jobs it does — cannot be traced: this lesson's station text is only in the rendered image, so there is nothing to match
- **nw-l05-o1** understand what the internet actually is — no station's text matches it
- **nw-l10-o1** know the basics of how cryptography can work with a simple key — nothing in this unit assesses it

- **nw-l10-o2** know how wireless devices authenticate each other before communicating data — nothing in this unit assesses it

Assessed, but matching no outcome this unit states

Each of these is a real exam question on a worksheet in this unit. Either the outcome it tests is worded differently, or it is genuinely assessing something the unit does not claim to teach.

- nw-101-x2 Name one advantage of a network.
- nw-101-x3 Name one disadvantage of a network.
- nw-104-x1 State the purpose of a router.
- nw-105-x1 Describe what the internet is.
- nw-112-x3 Describe what HTTPS adds to HTTP and why it matters.
- nw-113-x1 Give three benefits of dividing networking into layers.
- nw-114-test-x5 The concept of layers

What could not be traced

The wall text of nw-102, nw-104, nw-107 is painted into the 360 image and held nowhere as text, so an outcome in those lessons can only be traced through its activities and its exam questions. Read the rows above for them with that in mind.

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